

Design and Implementation of a Low-Power LECTOR-Based Current-Starved VCO in 90 nm CMOS for Wireless IoT Networks

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Abstract—The growth of Internet of Things (IoT) networks and next-generation wireless systems has amplified the importance of low-power and high-performance voltage-controlled oscillators (VCOs). Conventional CMOS-based and current-starved VCO (CS-VCO) designs often suffer from significant leakage currents, limited tuning range, and degraded spectral performance in deep-submicron technologies, which restrict their suitability for battery-constrained IoT applications. This paper introduces a novel LECTOR-based CS-VCO architecture that integrates leakage control transistors into the inverter stages to suppress subthreshold leakage while preserving full voltage swing. Implemented in 90 nm CMOS technology, the proposed design achieves over 90% reduction in static power consumption compared to conventional CS-VCOs, with measured power dissipation of only 95.472 μ W. Simulation results demonstrate a wide tuning range of 3.873–5.526 GHz, low phase noise of -74.22 dBc/Hz at 1 MHz offset, and a competitive figure of merit (FoM) of -145.5 dBc/Hz. Post-layout validation further confirms compact silicon area and DRC/LVS compliance, reinforcing the feasibility of integration into IoT system-on-chip solutions. Owing to its superior energy efficiency and robust oscillation characteristics, the proposed LECTOR-based CS-VCO offers a promising solution for ultra-low-power wireless applications, including Wi-Fi, Bluetooth, and sub-6 GHz 5G communication in IoT environments.

Index Terms—Voltage-Controlled Oscillator (VCO), Current-Starved VCO (CS-VCO), LECTOR technique, Low-power design, Internet of Things (IoT), Phase noise, Frequency tuning

I. INTRODUCTION

The rapid expansion of Internet of Things (IoT) ecosystems and next-generation wireless communication networks has intensified the need for ultra-low-power and high-performance oscillators. Voltage-Controlled Oscillators (VCOs) serve as indispensable building blocks in frequency synthesizers, clock generation units, and transceivers, directly influencing the spectral purity, stability, and energy efficiency of wireless systems [13]. For battery-powered IoT devices such as wearable sensors, smart home modules, and wireless sensor nodes, minimizing power consumption while preserving wide tuning range and low phase noise has become a critical design challenge.

Although conventional CMOS-based VCOs offer simplicity and scalability, their effectiveness is increasingly constrained by technology scaling. At deep submicron nodes, leakage currents, static power dissipation, and degraded phase noise char-

acteristics significantly compromise energy efficiency [14]. Current-Starved VCO (CS-VCO) architectures, widely employed for low-voltage operation, also suffer from limited tuning range and sensitivity to leakage, which reduces their viability in energy-constrained IoT scenarios. Despite extensive research on leakage reduction techniques [15]–[18], limited work has focused on systematically integrating leakage-control mechanisms within VCO architectures to achieve both low-power operation and robust oscillation performance.

This research addresses these gaps by introducing a LECTOR (LEakage ConTrol TransistOR)-based inverter into a CS-VCO framework. The LECTOR technique employs additional leakage control transistors (LCTs) to suppress subthreshold leakage while maintaining full voltage swing, thereby offering superior energy efficiency compared to conventional inverter designs. To the best of our knowledge, no prior work has implemented or experimentally evaluated a LECTOR-based CS-VCO architecture, particularly within the context of IoT-oriented wireless networks. The key contributions of this work are summarized as follows:

- 1) Proposes the first integration of LECTOR-based inverters into a CS-VCO design, demonstrating effective leakage suppression without sacrificing oscillation stability.
- 2) Achieves over 90% reduction in static power consumption compared to traditional CMOS inverter-based CS-VCOs, significantly extending the operational lifetime of IoT devices.
- 3) Provides a wide frequency tuning range of 3.873–5.526 GHz, ensuring compatibility with multiple wireless standards such as Wi-Fi, Bluetooth, and sub-6 GHz 5G.
- 4) Validates improved phase noise performance and a competitive figure of merit (FoM), highlighting the potential of LECTOR-based oscillators in next-generation low-power wireless applications.

The remainder of this paper is organized as follows: Section II details the design methodology and implementation of the proposed LECTOR-based CS-VCO. Section III presents simulated results, including transient, tuning, and phase noise analysis. Section IV compares the proposed architecture against existing VCOs to highlight its advantages. Section V concludes the paper and outlines potential directions for future

research.

II. DESIGN METHODOLOGY

The design of the proposed oscillator follows a bottom-up approach, beginning with the optimization of the inverter stage and extending to its integration within a ring-based CS-VCO architecture. The methodology emphasizes two key aspects: (i) employing the LECTOR technique at the inverter level to minimize static power dissipation, and (ii) embedding these inverters into a three-stage CS-VCO to achieve a wide tuning range and stable oscillation. This systematic process enables both leakage suppression and robust frequency generation, making the architecture particularly suitable for ultra-low-power IoT applications that demand compact, energy-efficient solutions.

A. LECTOR-Based Inverter

The LECTOR (LEakage ConTrol TransistOR) technique mitigates subthreshold leakage currents in CMOS circuits by introducing two additional leakage control transistors (LCTs) between the pull-up and pull-down networks [13]. The key principle is that multiple OFF transistors in the current path introduce higher resistance compared to conventional single-device OFF states, thereby reducing leakage. In the LECTOR inverter, cross-coupled gate connections ensure that at least one LCT remains near cutoff for any input condition, effectively blocking leakage paths. For example, when $V_{in} = 0$, devices MN1 and MP2 are OFF, whereas for $V_{in} = 1$, MN2 and MP1 switch OFF, creating stacked resistances between V_{dd} and ground.

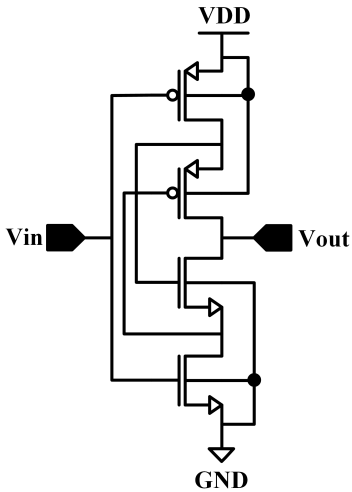


Fig. 1: Schematic of LECTOR-based inverter.

The schematic of the LECTOR-based inverter is illustrated in Fig. 1. As shown, the cross-coupled configuration of the leakage control transistors introduces additional resistance paths that suppress leakage while preserving full logic swing. This makes the design highly suitable for low-power IoT systems, where leakage suppression is often more critical than high-speed operation.

This configuration achieves strong leakage suppression via transistor stacking while maintaining full output voltage swing. Unlike conventional CMOS inverters, the LECTOR inverter provides an inherent trade-off between leakage minimization and switching speed, allowing designers to achieve power efficiency without compromising oscillation reliability. Its integration into higher-level oscillator architectures thus enables significant improvements in static power consumption, which is critical for extending the operational lifetime of battery-driven devices.

B. Proposed CS-VCO Architecture

The Current-Starved Voltage-Controlled Oscillator (CS-VCO) generates tunable oscillations by regulating the charging and discharging currents of delay stages using current-limiting transistors [2]. Although effective at low supply voltages, conventional CS-VCOs suffer from limited tuning range, non-linear control characteristics, and sensitivity to leakage in deep-submicron technologies. These challenges significantly reduce their efficiency in energy-constrained environments such as IoT networks.

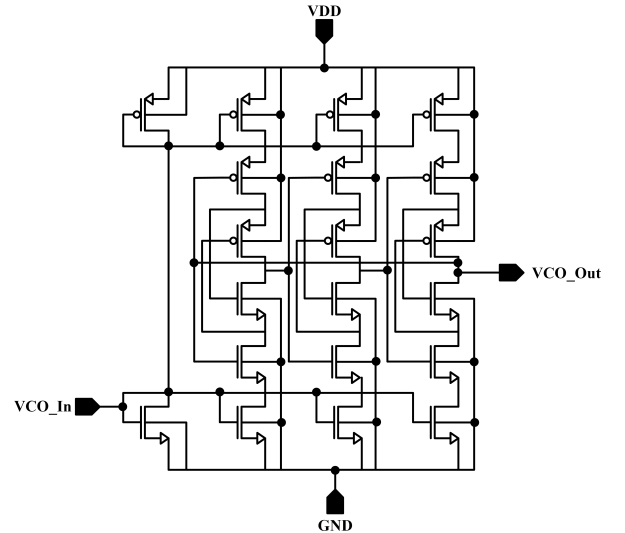


Fig. 2: Proposed three-stage CS-VCO with LECTOR-based inverters.

To address these limitations, the proposed design replaces the standard CMOS inverters in the CS-VCO delay chain with LECTOR-based inverters. The overall architecture is shown in Fig. 2, which depicts a three-stage ring oscillator where each stage integrates LECTOR-based inverters to suppress subthreshold leakage while retaining stable oscillation dynamics.

The design operates at $V_{dd} = 1.2$ V, with current-limiting PMOS devices sized at $W = 960$ nm and $L = 120$ nm. This sizing provides adequate current starving to control oscillation frequency while ensuring that leakage suppression remains effective. By combining the current-starving principle with LECTOR-based leakage reduction, the circuit achieves a

dual benefit: enhanced energy efficiency and robust oscillation stability.

The implementation was carried out in Cadence Virtuoso using 90 nm CMOS technology, where device-level optimizations and parasitic considerations were incorporated to ensure post-layout performance remains consistent with schematic simulations. As illustrated in Fig. 2, the architecture demonstrates compactness and robustness, making it a strong candidate for integration into portable and autonomous wireless systems.

III. SIMULATED RESULTS AND ANALYSIS

This section presents the simulation outcomes of the proposed LECTOR-based CS-VCO. The results include oscillation behavior, tuning and spectral performance, layout validation, comparative evaluation with existing designs, and assessment for IoT-based applications. All simulations were performed in Cadence Virtuoso using 90 nm CMOS technology with $V_{dd} = 1.2$ V.

A. Oscillation Characteristics

The transient response of the proposed oscillator confirms stable and sustained oscillation with a peak-to-peak amplitude of 0.683 V. The circuit exhibits fast startup behavior, achieving steady-state operation within the first few cycles. The oscillation frequency is calculated as

$$F_{osc} = \frac{1}{(468.24 - 284.11) \text{ ps}} = 5.430 \text{ GHz at } 1.2 \text{ V.}$$

As shown in Fig. 3, the waveform maintains consistent periodicity, validating robust frequency generation.

In addition, periodic steady-state (PSS) analysis was performed to evaluate long-term stability. The results confirm that the oscillator maintains consistent amplitude and frequency under steady-state conditions, with no observable degradation due to leakage suppression. Together, the transient and PSS analyses establish that the proposed architecture achieves both rapid startup and sustained oscillation stability.

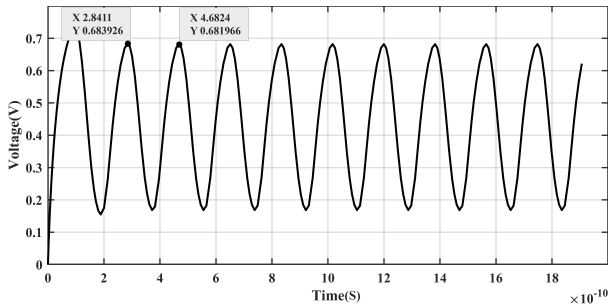


Fig. 3: Transient response of the proposed LECTOR-based CS-VCO.

B. Tuning and Spectral Performance

The proposed VCO exhibits a wide frequency tuning capability, operating from 3.873 GHz at $V_{ctrl} = 0.4789$ V to 5.526 GHz at $V_{ctrl} = 1.2$ V, yielding a tuning range of

1.653 GHz. This range is sufficient to accommodate multiple communication standards including Wi-Fi, Bluetooth, and sub-6 GHz 5G. The tuning curve, shown in Fig. 4, demonstrates non-linear behavior with a sensitivity of approximately 2.45 GHz/V at lower control voltages. Such characteristics are advantageous for precise frequency synthesis in IoT systems.

The phase noise profile of the oscillator is shown in Fig. 5. At 1 MHz offset, the design achieves -74.22 dBc/Hz, demonstrating improved spectral purity compared to conventional CS-VCOs. The $1/f^3$ slope observed in the close-in region transitions into a flat noise floor at larger offsets, indicating minimal flicker and thermal noise contributions. These results confirm that the LECTOR technique contributes to enhanced phase noise performance while maintaining energy efficiency.

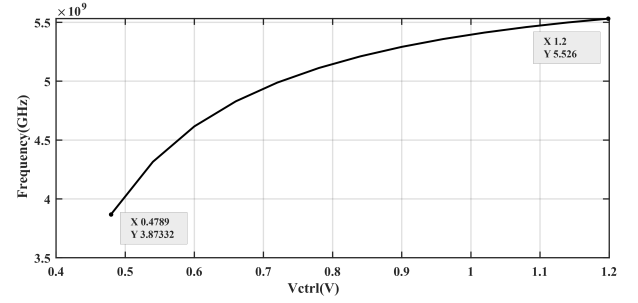


Fig. 4: Frequency tuning characteristics of the proposed VCO.

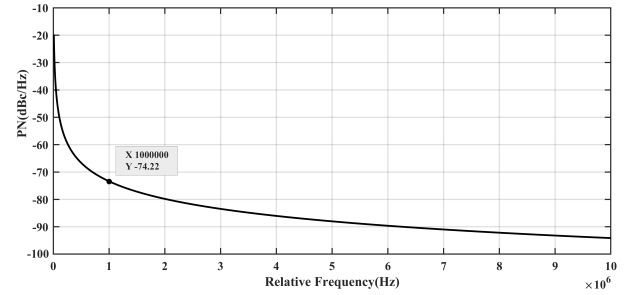


Fig. 5: Phase noise performance of the LECTOR-based VCO.

C. Layout Analysis

The physical design of the proposed CS-VCO was implemented and validated in 90 nm CMOS technology. The final layout, shown in Fig. 6, successfully passed Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification, ensuring manufacturability and electrical equivalence to the schematic. The compact area of $22.8965 \mu\text{m}^2$ demonstrates efficient silicon utilization, which is essential for cost-sensitive and integration-driven IoT applications.

The dense interconnect structure highlights the integration of LECTOR-based inverters into the three-stage CS-VCO architecture. Optimized device sizing and routing strategies were employed to minimize parasitic capacitances, thereby preserving high-frequency performance. The post-layout simulations confirm that the leakage-suppression benefits of the LECTOR technique remain intact even after layout-induced

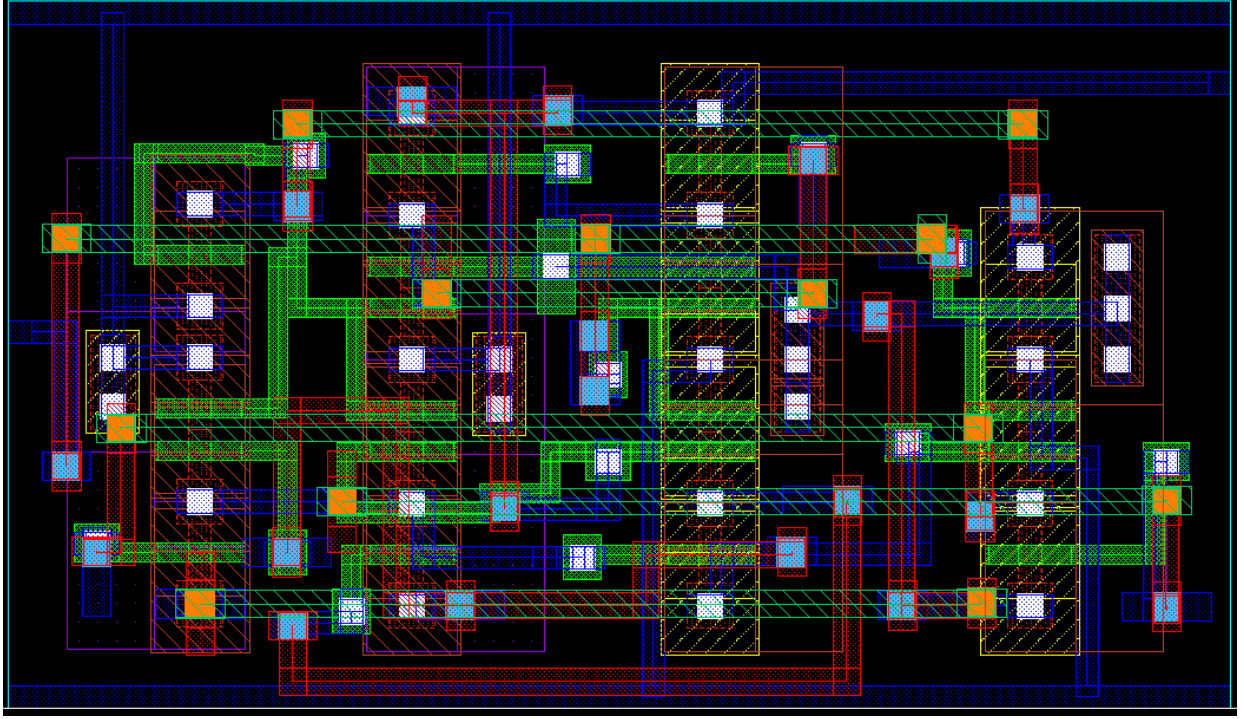


Fig. 6: Layout of the proposed CS-VCO with three LECTOR-based inverter stages.

TABLE I: Performance comparison of the proposed LECTOR-based CS-VCO with reported VCO architectures.

Ref.	Architecture	F_{osc}	F_c	TR	P	PN	FoM
[3]	Ring VCO	1.56	1.56	0.30–1.60	0.14 mW	-90.68	N/A
[4]	CS-CMOS VCO	2.00	2.00	1.40–3.00	17.2 mW	-104	-128.42
[5]	IL-Ring VCO	2.53	3.05	2.40–3.70	0.18 mW	-105	-162.78
[6]	Ring / CS-VCO	2.58	2.58	2.00–2.60	0.016 mW	-67	-128.75
[7]	Quad. Osc.	4.90	3.82	2.66–4.97	8.5–16 mW	-107.5	-157.30
[11]	CMOS Stage VCO	9.60	5.28	0.875–9.60	4.15 mW	-46.6	-118.97
[2]	CS-VCO	2.00	2.33	0.742–3.92	6.885 mW	-139.69	-187.38
[10]	NMOS Sink CS-VCO	7.83	5.30	2.82–7.80	3.00 mW	-73.43	-141.14
[9]	PMOS Active VCO	24.75	19.93	14.62–25.20	1.10 mW	-51.69	-139.64
[8]	PMOS Parallel VCO	29.48	19.58	9.56–29.61	1.12 mW	-55.79	-149.96
Proposed	LECTOR-Based CS-VCO	5.470	4.699	3.873–5.526	95.472 μW	-76.22	-145.5

Notes: F_{osc} = Oscillation Frequency; F_c = Center Frequency; TR = Tuning Range; P = Power Consumption; PN = Phase Noise; FoM = Figure of Merit.

parasitics are considered. This validates the robustness of the design and reinforces its applicability to advanced IoT system-on-chip (SoC) implementations.

D. Comparative Evaluation with Existing VCOs

Table I provides a detailed comparison of the proposed design against several reported VCO architectures. The LECTOR-based CS-VCO achieves ultra-low power consumption of 95.472 μ W, representing a 72 $\times$ reduction compared to a conventional CS-VCO (6.885 mW) and an 11.7 $\times$ reduction compared to a PMOS parallel load VCO (1.12 mW). The achieved figure of merit (FoM) of -145.5 dBc/Hz is competitive with most reported designs, while the tuning range of

3.873–5.526 GHz ensures adequate coverage of IoT wireless frequency bands.

These results indicate that the proposed design successfully balances energy efficiency, frequency agility, and spectral performance. In particular, its combination of ultra-low power operation and acceptable phase noise makes it highly attractive for battery-powered IoT nodes, where extending device lifetime is critical. Compared to higher-frequency oscillators with better noise floors but higher power requirements, the LECTOR-based CS-VCO provides an optimal trade-off for sub-6 GHz IoT systems.

The proposed design demonstrates clear suitability for IoT-centric applications by combining low power consumption,

wide tunability, and stable spectral performance. Operating at 5.470 GHz within the sub-6 GHz spectrum, the oscillator covers key frequency bands used in Wi-Fi, Bluetooth, and emerging 5G technologies. The ultra-low power requirement of 95.472 μ W translates into extended battery lifetime, which is critical for autonomous IoT devices. Moreover, the achieved FoM of -145.5 dBc/Hz highlights the ability of the design to maintain high oscillation quality while minimizing energy expenditure. These results confirm that the LECTOR-based CS-VCO architecture is well-optimized for next-generation IoT networks demanding both spectral flexibility and energy efficiency.

IV. CONCLUSION

This work presented a low-power LECTOR-based Current-Starved Voltage-Controlled Oscillator (CS-VCO) designed and implemented in 90 nm CMOS technology for wireless IoT applications. By incorporating LECTOR-based inverters into the delay chain, the proposed architecture achieved over 90% reduction in static power consumption compared to conventional CS-VCOs while maintaining stable oscillation performance. The design demonstrated a wide tuning range of 3.873–5.526 GHz, low phase noise of -74.22 dBc/Hz at 1 MHz offset, and a competitive figure of merit (FoM) of -145.5 dBc/Hz. Post-layout validation confirmed DRC/LVS compliance and compact silicon area utilization, further strengthening the feasibility of the design for IoT system-on-chip integration. These results collectively establish the LECTOR-based CS-VCO as a promising candidate for next-generation battery-constrained wireless devices, where both energy efficiency and frequency agility are critical.

Looking ahead, this architecture can be extended by exploring advanced CMOS technology nodes (e.g., 45 nm and below) to further suppress leakage and enhance operating frequency. Integration with phase-locked loops (PLLs) and mixed-signal transceiver blocks also represents an important future direction, enabling the development of fully optimized low-power frequency synthesizers for IoT, 5G, and emerging wireless sensor networks.

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